

Memory-Centric Agentic Inference — cycles 1-3 clone 2

2026-05-11

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Abstract

Clone 2 designed and validated the semantic-cache and durable-state risk measurement branch for M-EXP-1, the deferred-constant measurement harness. Its required artifact is `memory-centric-agentic/experiments/cache_durable_risk_m`

The branch covers two deferred constants from the parent synthesis:

- DC-004: semantic-cache correctness and invalidation cost.
- DC-003: remote durable object-store latency distributions for agent state.

The main result is a measurement design that can downgrade architecture choices instead of only reporting cache or storage performance. Option B, the memory-object-aware runtime, is preserved for retrieval-augmented generation only when semantic-cache hits remain valid after freshness, provenance, tenant/cache-salt, poisoning, validation, and recovery checks. Option C, the trajectory/DAG-aware memory fabric, is preserved only when durable replay p50/p95/p99 tails, dependency fan-in, pointer validity, consistency waits, retention state, and replay recovery remain below retained-value margins.

The branch was validated by audit. The remaining issue is explicitly parent-scope: the parent measurement CSVs exist as managed artifacts but are not yet registered cleanly in the promise ledger, and the clone-2 CDR rows still need parent M-EXP-1 integration.

Introduction

The root project asks whether future agentic LLM inference systems should be designed around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Earlier parent work produced three architecture options:

- **Option A:** conventional request/model/KV-centric serving.
- **Option B:** a memory-object-aware runtime that tracks reusable objects such as retrieved context, prefix cache entries, semantic cache entries, tool outputs, provenance pointers, and invalidation state.

- **Option C:** a trajectory/DAG-aware memory fabric that treats branch state, verifier state, durable workspace state, and replay dependencies as first-class scheduling and placement inputs.

The final synthesis identified several deferred constants: quantities that could reverse architecture conclusions if measured values are unfavorable. Clone 2 focused on two of them.

DC-004 asks whether semantic caches and related approximate reuse mechanisms actually provide safe retained value. A semantic cache stores or retrieves prior responses or objects by similarity rather than exact identity; this creates risk from false positives, stale source versions, poisoned entries, tenant isolation failures, and expensive recovery. Public work exists on prefix caching and semantic caching mechanisms, but the parent synthesis kept correctness and invalidation cost deferred rather than treating raw hit rate as sufficient evidence [12], [13].

DC-003 asks whether durable object-store replay is fast and reliable enough for agentic state. Durable replay means reading prior workspace objects, checkpoints, tool outputs, summary-pointer state, verifier evidence, or trajectory logs so an agent can resume or merge work. The relevant quantity is not average storage bandwidth. The risk is tail latency: p95 and p99 replay delays can dominate verifier or merge paths, especially when a replay path depends on many objects. NVMe and storage-interface sources provide capability context, but the agentic replay distribution remains a deferred measurement [6].

Methodology

This report consolidates clone-2 cycles 1-3. It reports completed work and audit outcomes; it does not re-audit the artifact.

The source sessions supplied for this report were:

	Cycle	Role	Session ID	Source content
	1	researcher	9b8a5162-0640-43a8-a6b4-584b33146f6d	Session 146609: this session describes sibling clone 0 trajectory-reuse output, not clone 2. It is useful only as fanout context.
	1	worker	11dd4699-c391-40a6-95cd-5856c35f1f8c	Verified and finalized the clone-2 cache/durable risk plan, reported 291 lines, and wrote a merge report.
	1	auditor	6ec8418d-8aa6-4db0-bd58-fad4d8eab07a	Found one moderate integration ambiguity: missing per-experiment instrumentation mappings. Validated the patched 308-line artifact.
	2	researcher	c4395aeb-ac9e-4823-8672-169174033372	Rebuilt clone 2 as validated parent-integration input and specified how parent CSVs should consume it.
	2	worker	7363fc26-9023-45a6-8d13-1361f84e044e	Validated artifact and reported a ledger event UUID later found absent.

Cycle	Role	Session ID	Source content
2	auditor	eb0ead1b-2df3-4f64-8118-V881b00303395	clone-2 artifact and noted the claimed ledger event was not present in <code>promise_ledger.jsonl</code> .
3	researcher	23078c5d-affe-4b80-8b96-D152cd86eb3d	Done 2 should not be reopened; parent integration is the next action.
3	worker	83b41bdd-a464-4283-971e-Pad1529fe1abd	Parent integration build work and preserved the validated branch as parent-integration input.
3	auditor	04dd614a-8fdf-4b00-8ed3-V11ab3c89e7026	Validated all experiment IDs, thresholds, and required markers remain present; documented parent CSV orphan warnings.

The reporting pass also used the branch artifact, the supplied audit report, `REFERENCES.md`, `MANIFEST.md`, `promise_ledger.jsonl`, and the current parent measurement CSVs.

Results

Cycle 1: Measurement Plan and Audit Patch

Cycle 1 produced the required clone-2 artifact:

`memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md`

The worker session `11dd4699-c391-40a6-95cd-856c35ff8cf8` reported that the plan covered:

- semantic false positives,
- stale or poisoned semantic-cache hits,
- invalidation and recovery cost,
- durable object-store replay tails,
- p50/p95/p99 durable latency,
- Option B downgrade logic,
- Option C downgrade logic,
- experiment IDs and threshold IDs.

The cycle 1 audit session `6ec8418d-8aa6-4db0-bd58-ad4d8deb07a3` found one moderate defect and fixed it. The defect was not a missing concept; it was an integration ambiguity. The plan listed branch-level instrumentation fields, but it did not map required `instrumentation_fields` per experiment row even though the parent harness schema requires that field in `measurement_experiment_specs.csv`.

The auditor added a per-experiment instrumentation map for all eleven experiments:

- CDR-SEM-001 through CDR-SEM-006,
- CDR-DUR-001 through CDR-DUR-005.

After the patch, the artifact was 308 lines. The audit reported:

- critical issues: none,
- moderate issues: one found and fixed,
- minor issues: none,
- `promise_check`: green,
- `org_check`: green,

- decision: validated.

Cycle 2: Parent Integration Framing and Ledger Warning

Cycle 2 shifted the branch from design work to parent integration readiness. The researcher session `c4395aeb-ac9e-4823-8672-1e9171033372` stated that the clone-2 plan should feed five parent harness tables:

- `data/measurement_experiment_specs.csv`,
- `data/measurement_required_fields.csv`,
- `data/measurement_thresholds.csv`,
- `data/measurement_claim_update_matrix.csv`,
- `data/measurement_synthetic_probe_results.csv`.

The worker session `7363fc26-9023-45a6-8d13-136f88480448` revalidated the artifact and reported a ledger event UUID:

`523f4e76-04d4-4002-8000-000000000002`

The cycle 2 audit session `eb0ead1b-2df3-4f64-8118-881903303395` found that this UUID was absent from `promise_ledger.jsonl`. The auditor did not append a replacement event because clone 2 validates a branch design, not the full parent M-EXP-1 milestone. The audit classified the issue as moderate audit-trail inconsistency, not an artifact defect.

The audit validated the branch content and confirmed:

- all 11 expected experiment IDs are present,
- all 10 expected threshold IDs are present,
- semantic correctness and durable replay risks remain separate,
- parent integration readiness is met for this branch,
- the absent ledger event should not be relied on downstream.

Cycle 3: Stability Check and Parent-Scope Warning

Cycle 3 made no new design changes. The researcher session `23078c5d-affe-4b80-8b96-152cd86ecb38` explicitly instructed that clone 2 should be treated as validated parent-integration input, not reopened.

The worker session `83b41bdd-a464-4283-971e-adb529fe1abd` therefore performed no additional build work. It preserved the existing artifact and restated the validated contents:

- 11 experiment specs,
- 10 downgrade thresholds,
- DC-004 coverage for semantic false positives, stale/poisoned hits, invalidation, recovery, tenant/cache-salt safety, and valid-hit retained value,
- DC-003 coverage for durable replay p50/p95/p99 tails, dependency fan-in, pointer validity, retention gates, and replay-failure risk.

The final audit session `04dd614a-8fdf-4b00-8ed3-1aa6c89e7026` validated the stable artifact. It also reported a parent-scope warning: `promise_check` exits successfully but reports orphan managed artifacts for the parent measurement CSVs. The warning covers:

- `data/measurement_claim_update_matrix.csv`,
- `data/measurement_experiment_specs.csv`,
- `data/measurement_required_fields.csv`,
- `data/measurement_synthetic_probe_results.csv`,
- `data/measurement_thresholds.csv`.

The audit decision remained validated because this warning is not a clone-2 artifact defect. It must be resolved or explicitly documented during parent M-EXP-1 integration.

Measurement Design

DC-004: Semantic-Cache Risk

The plan's first mechanism hypothesis is that semantic-cache benefit is gated by valid-hit rate, not raw-hit rate. A raw hit means the cache returned something similar. A valid hit means the returned object is also fresh, trusted, tenant-safe, provenance-valid, and recoverable.

The plan defines safe semantic retained value as:

```
V_semantic_safe =  
  P_hit * P_valid_given_hit * C_recompute_avoided  
  - C_lookup  
  - C_validation  
  - C_invalidation  
  - C_recovery  
  - E[C_wrong_reuse]
```

Option B remains justified for RAG only if this value is positive after registry, policy, and migration overheads. The plan makes several hard-gate cases explicit:

Condition	Decision
P_hit = 0	Semantic cache cannot justify Option B.
P_valid_given_hit = 0	Reuse must be disabled or exact provenance pointers required.
invalidation_signal != none	Reuse value is forced nonpositive until revalidated or recomputed.
tenant_scope or cache_salt mismatch	Reuse is forbidden independent of hit quality.
Missing provenance_id or stale source_version	Reuse is blocked or falls back to exact retrieval/recompute.

The six semantic-cache experiments are:

Experiment ID	Measurement
CDR-SEM-001	raw hit rate versus valid-hit rate
CDR-SEM-002	semantic false-positive rate by match-score bucket
CDR-SEM-003	stale-hit rate after source-version and invalidation changes
CDR-SEM-004	safe hit rate after tenant-scope and cache-salt enforcement
CDR-SEM-005	recovery latency and recovery action after invalid cache hit
CDR-SEM-006	poisoning or untrusted-provenance rejection rate

The semantic-cache downgrade thresholds are:

Threshold ID	Decision
T-SEM-VALID-001	Disable semantic-cache retained-value credit when valid-hit value cannot pay lookup, validation, recovery, and wrong-reuse costs.
T-SEM-STALE-002	Require exact provenance or revalidation when stale-hit loss erases recompute savings.
T-SEM-ISOLATION-003	Downgrade Option B if safe hit rate collapses after tenant/cache-salt gates.
T-SEM-POISON-004	Forbid correctness-sensitive reuse when untrusted reuse rate is nonzero without trusted provenance.

Threshold ID	Decision
T-SEM-RECOVERY-005	Treat cache lookup as overhead when lookup, validation, and recovery cost exceed exact retrieval or recompute.

DC-003: Durable-State Replay Risk

The plan's second mechanism hypothesis is that durable replay benefit is gated by tail latency and dependency width. Durable storage can preserve workspace, trajectory, checkpoint, tool-output, verifier, and branch state. That preservation only supports Option C if replay does not dominate the saved recomputation or coordination time.

The plan defines durable replay value as:

```
V_durable_replay =
  P_replay * C_recompute_avoided
  - L_store_read(p)
  - L_consistency(p)
  - C_pointer_validation
  - C_reconstruction
  - E[C_replay_failure]
```

The latency percentile p must include at least p50, p95, and p99. This is required because agentic replay can block on the slowest object in a dependency path.

The plan makes these special cases explicit:

Condition	Decision
$P_{\text{replay}} = 0$	Durable storage remains audit/logging only; it creates no retained execution value.
$L_{\text{store_read}}(p99)$ is unbounded or too high	Latency-sensitive replay is invalid; use local pinning, recompute, or Option B fallback.
$\text{dependency_width} = 1$	Tail amplification is minimal, but p95/p99 still matter.
Large dependency_width	End-to-end replay approaches the maximum of many object reads; p99 can dominate.
Expired $\text{durability_horizon}$ without active $\text{retention_hold_state}$	Reuse is a retention violation, not a performance optimization.
Missing recovery pointer	Summary-plus-pointer replay must fall back to exact retained state or recompute.

The five durable-state experiments are:

Experiment ID	Measurement
CDR-DUR-001	durable workspace write/read p50/p95/p99 by size and consistency mode
CDR-DUR-002	dependency-path max replay latency
CDR-DUR-003	verifier/tool-output summary-pointer recovery latency
CDR-DUR-004	retention-horizon and hold-state violations
CDR-DUR-005	replay success rate under object-store tail injection

The durable-state downgrade thresholds are:

Threshold ID	Decision
T-DUR-TAIL-001	Avoid remote durable replay when p95 replay latency exceeds saved recomputation time.

Threshold ID	Decision
T-DUR-P99-002	Downgrade Option C when p99 max-path latency exceeds verifier or merge deadline.
T-DUR-FANIN-003	Downgrade when dependency fan-in tail amplification erases retained-value margin.
T-DUR-RECOVERY-004	Disable summary-plus-pointer when pointer is invalid or reconstruction is slower than recompute.
T-DUR-RETENTION-005	Forbid reuse when durability horizon has expired and no retention hold is active.

Integration Status

The clone-2 plan is complete as a branch artifact. It is not yet complete as parent M-EXP-1 CSV integration.

The plan tells the parent harness to add:

- 11 experiment rows: CDR-SEM-001 through CDR-SEM-006, and CDR-DUR-001 through CDR-DUR-005;
- 10 threshold rows: T-SEM-VALID-001 through T-SEM-RECOVERY-005, and T-DUR-TAIL-001 through T-DUR-RETENTION-005;
- required fields such as `cache_key_hash`, `tenant_scope`, `cache_salt`, `semantic_match_score`, `correctness_label`, `recovery_latency_ms`, `durable_object_id`, `dependency_path_id`, `dependency_width`, `read_latency_ms`, `consistency_wait_ms`, `pointer_valid`, and `retention_hold_state`;
- claim-update mappings for CL-002, CL-003, CL-006, CL-008, CL-009, and CL-010;
- synthetic probe rows for low/high false-positive, stale-hit, recovery-cost, durable-tail, fan-in, pointer-validity, and retention-expiry settings.

The current parent measurement CSVs in the workspace are populated for DC-005 trajectory reuse, not yet for clone-2 CDR-* rows. This matches the audit guidance: clone 2 is a validated input, while parent integration remains the next meaningful target.

Discussion

Clone 2 narrows the meaning of “reuse” in the memory-centric architecture. Reuse is not counted just because a cache lookup hits or a durable object can be read. Reuse counts only when it survives correctness, freshness, provenance, isolation, recovery, latency, and retention gates.

For Option B, this prevents semantic cache hit rate from becoming a misleading metric. A high raw hit rate can still downgrade to Option A if valid-hit rate is low, if false positives persist in high-score buckets, if stale entries reach correctness-sensitive paths, if tenant/cache-salt enforcement collapses safe reuse, or if recovery cost exceeds exact retrieval.

For Option C, this prevents durable storage from being treated as automatically useful. Durable state can support branch/merge replay, verifier reuse, and workspace continuation only when p95/p99 replay tails and fan-in amplification stay below retained-value margins. If pointer recovery is invalid, retention state is expired, or max-path p99 exceeds verifier or merge deadlines, durable replay should downgrade to local retention, Option B object runtime, recompute, or forbidden reuse.

The plan also keeps sibling scopes separate. DC-003 is durable replay-tail risk. DC-004 is semantic-cache correctness, invalidation, and recovery risk. DC-005 trajectory reuse distribution and DC-006 provenance-validation overhead are related, but they should not be collapsed into the same benchmark. The auditors repeatedly emphasized this boundary because collapsing the deferred constants would turn a discriminating measurement plan into a generic cache/storage test.

Conclusions and Recommendations

Clone 2 completed its scoped objective. The validated artifact `memory-centric-agent/experiments/cache_durable_risk_me` gives the parent M-EXP-1 harness a concrete design for measuring DC-004 and DC-003.

The central conclusions are:

1. Option B RAG should be credited for semantic-cache reuse only when valid semantic retained value remains positive after correctness, freshness, provenance, tenant/cache-salt, poisoning, validation, recovery, and wrong-reuse costs.
2. Option C durable replay should be credited only when durable p50/p95/p99 replay tails, dependency fan-in, consistency waits, pointer reconstruction, retention validity, and replay-failure risk remain below retained-value margins.
3. Unsafe reuse is a hard block, not a cost tradeoff, for tenant/cache-salt mismatch, missing provenance in correctness-sensitive paths, stale source versions, untrusted provenance, invalid pointers, and expired retention without active hold.
4. Clone 2 should now be integrated into the parent measurement CSVs rather than reopened for further design.

The next parent action is to ingest the plan into the M-EXP-1 CSVs while preserving the distinction between DC-004, DC-003, DC-005, and DC-006, then resolve or explicitly document the parent CSV orphan warnings before full M-EXP-1 validation.

References

- [6] NVM Express, “Specifications,” NVM Express, accessed 2026-05-11. <https://nvmexpress.org/specifications/>
- [12] vLLM Project, “Automatic Prefix Caching,” vLLM Documentation, accessed 2026-05-11. https://docs.vllm.ai/en/latest/design/prefix_caching/
- [13] Sajal Regmi and Chetan Phakami Pun, “GPT Semantic Cache: Reducing LLM Costs and Latency via Semantic Embedding Caching,” arXiv, 2024. <https://arxiv.org/abs/2411.05276>

Appendix: Implementation Details

Code Organization

The required clone-2 artifact is:

File	Lines	Purpose
<code>memory-centric-agentic/experiments/cache_durable_risk_measurement_plan</code>	308	Measurement design for DC-004 semantic-cache correctness/invalidation risk and DC-003 durable replay-tail risk.

Related sibling experiment plans are:

File	Lines	Purpose
<code>memory-centric-agentic/experiments/trajectory_reuse_measurement_plan</code>	218	Measurement design for DC-005 trajectory reuse.
<code>memory-centric-agentic/experiments/provenance_overhead_measurement_plan</code>	114	Measurement design for DC-006 provenance-validation overhead.

The parent measurement CSVs currently present in the workspace are:

File	Lines	Current status
<code>data/measurement_experiment_specs.csv</code>	8	Contains DC-005 rows; clone-2 CDR rows remain parent integration input.
<code>data/measurement_required_fields.csv</code>	14	Contains DC-005 fields; clone-2 fields remain parent integration input.

File	Lines	Current status
data/measurement_thresholds.csv	5	Contains DC-005 thresholds; clone-2 thresholds remain parent integration input.
data/measurement_claim_update_matrix.csv	12	Contains DC-005 claim updates; clone-2 claim updates remain parent integration input.
data/measurement_synthetic_probe_results.csv	7	Contains DC-005 synthetic probe rows; clone-2 probe rows remain parent integration input.

No figures were created by clone 2. The deliverable is a markdown measurement plan.

Test Results

The supplied audit report and session audits recorded these validation results:

Check	Result
Required artifact exists and is nonempty	passed
Artifact line count	308
CDR-SEM-001 through CDR-SEM-006 present	passed
CDR-DUR-001 through CDR-DUR-005 present	passed
T-SEM-VALID-001 through T-SEM-RECOVERY-005 present	passed
T-DUR-TAIL-001 through T-DUR-RETENTION-005 present	passed
DC-003/DC-004 markers present	passed
valid-hit gating present	passed
p50/p95/p99 durable tails present	passed
tenant/cache-salt, pointer validity, retention state, negative tests, and claim update map present	passed
org_check	green
promise_check	exits 0 with parent-scope orphan warnings

During this reporting pass, MANIFEST.md was updated to include the clone-2 experiment plan and the current parent integration status. `python3 -m long_exposure.tools.org_check <workspace>` was green. `python3 -m long_exposure.tools.promise_check <workspace>` exited successfully and reported the parent measurement CSV orphan warnings already noted by audit, plus clone-0 rendered report artifacts.

Session References

	Cycle	Researcher	Worker	Auditor
	1	9b8a5162-0640-43a8-a6b418403690ff80d	18403690ff80d-40a6-95cd6858438df8a16	18403690ff80d-4db0-bd58-ad4d8de
	2	c4395aeb-ac9e-4823-86727369f72039073	18403690ff80d-45a6-8d13eb06a88080448	18403690ff80d-4f64-8118-8819033
	3	23078c5d-affe-4b80-8b9683521b886ea468	18403690ff80d-4283-971e04dd629fe31dd	18403690ff80d-4b00-8ed3-1aa6c89

Cross-Reference Map

Origin	Consuming artifact	Flow
data/runtime_ablation_results.csv	cache_durable_risk_measurement_plan	Defines the Option B and Option C collapse mechanisms that the plan measures.
data/security_invalid_trace_fixtures	cache_durable_risk_measurement_plan	Supplies negative-test patterns for missing provenance, tenant/cache-salt mismatch, invalid pointer, and retention failures.
data/calibration_deferred_constants	cache_durable_risk_measurement_plan	Provides DC-003 and DC-004 as deferred constants to turn into measurement designs.
cache_durable_risk_measurement_plan	measurement_experiment_specs.csv	Should add CDR-SEM-* and CDR-DUR-* experiment rows with per-experiment instrumentation_fields.
cache_durable_risk_measurement_plan	measurement_required_fields.csv	Should add semantic-cache and durable-state required fields.
cache_durable_risk_measurement_plan	measurement_thresholds.csv	Should add T-SEM-* and T-DUR-* downgrade thresholds.
cache_durable_risk_measurement_plan	measurement_claim_update_matrix.csv	Should update claims CL-002, CL-003, CL-006, CL-008, CL-009, and CL-010.
cache_durable_risk_measurement_plan	measurement_synthetic_probe_results	Should add synthetic probes for valid hit collapse, stale/poisoned hits, recovery cost, durable tails, fan-in, pointer validity, and retention expiry.